

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
db={"name":["Raj","sneha","Sona","lekshmi","joey","Alice","Bob","Charlie",
"David","Neha","Rahul","Ajay","ram","sham","sree","Gokul","Maya","zoe","suma","Beena","Hari"],
"age":[25,32,22,21,44,32,31,22,25,32,22,21,44,32,31,22,31,33,22,45,36],
"Depart":
["hr","sales","finance","marketing","hr","sales","finance","marketing","finance","hr","hr","sales","marketi
ng","sales","admin","sales","it","admin","it","sales","finance"],
"Salary":
[20000,18000,31000,6000,10000,11000,70000,22000,10000,20000,20000,18000,31000,6000,1
0000,11000,70000,22000,10000,20000,17000],
"Experience":[3,5,2,3,1,8,5,6,2,7,3,5,2,3,1,8,5,6,2,7,3],
"Performance scr":[6,7,4,9,7,8,6,5,4,8,6,7,4,9,7,8,6,5,4,8,4]}
df=pd.DataFrame(db)
print(df)
```

	name	age	Depart	Salary	Experience	Performance scr
0	Raj	25	hr	20000	3	6
1	sneha	32	sales	18000	5	7
2	Sona	22	finance	31000	2	4
3	lekshmi	21	marketing	6000	3	9
4	joey	44	hr	10000	1	7
5	Alice	32	sales	11000	8	8
6	Bob	31	finance	70000	5	6
7	Charlie	22	marketing	22000	6	5
8	David	25	finance	10000	2	4
9	Neha	32	hr	20000	7	8
10	Rahul	22	hr	20000	3	6
11	Ajay	21	sales	18000	5	7
12	ram	44	marketing	31000	2	4
13	sham	32	sales	6000	3	9
14	sree	31	admin	10000	1	7
15	Gokul	22	sales	11000	8	8
16	Maya	31	it	70000	5	6
17	zoe	33	admin	22000	6	5
18	suma	22	it	10000	2	4
19	Beena	45	sales	20000	7	8
20	Hari	36	finance	17000	3	4

```
Salary_array=df['Salary'].to_numpy()
Experience_array=df['Experience'].to_numpy()
print(Salary_array)
print(Experience_array)
```

```
[20000 18000 31000 6000 10000 11000 70000 22000 10000 20000 20000 18000
31000 6000 10000 11000 70000 22000 10000 20000 17000]
[3 5 2 3 1 8 5 6 2 7 3 5 2 3 1 8 5 6 2 7 3]
```

```
Salary=np.array([20000,18000,31000,6000,10000,11000,70000,22000,10000,20000,20000,180
```

```
00,31000,6000,10000,11000,70000,22000,10000,20000,17000])
mean=np.mean(Salary)
median=np.median(Salary)
std_deviation=Salary.std()
print("mean:",mean)
print("median:",median)
print("std_deviation:",std_deviation)
```

```
mean: 21571.428571428572
median: 18000.0
std_deviation: 17139.682245683354
```

```
Experience=np.array([3,5,2,3,1,8,5,6,2,7,3,5,2,3,1,8,5,6,2,7,3])
min_exp=np.min(Experience)
max_exp=np.max(Experience)
print("min_exp:",min_exp)
print("max_exp:",max_exp)
```

```
min_exp: 1
max_exp: 8
```

```
salary=np.array([20000,18000,31000,6000,10000,11000,70000,22000,10000,20000,20000,18000,31000,6000,10000,11000,70000,22000,10000,20000,17000])
min_sal=np.min(Salary)
max_sal=np.max(Salary)
normalise=(Salary-min_sal)/(max_sal-min_sal)
print("normalise:",normalise)
```

```
normalise: [0.21875 0.1875 0.390625 0. 0.0625 0.078125 1. 0.25
0.0625 0.21875 0.21875 0.1875 0.390625 0. 0.0625 0.078125
1. 0.25 0.0625 0.21875 0.171875]
```

```
df.head(5)
```

	name	age	Depart	Salary	Experience	Performance scr
0	Raj	25	hr	20000	3	6
1	sneha	32	sales	18000	5	7
2	Sona	22	finance	31000	2	4
3	lekshmi	21	marketing	6000	3	9
4	joey	44	hr	10000	1	7

```
df.describe()
```

	age	Salary	Experience	Performance scr
count	21.000000	21.000000	21.000000	21.000000
mean	29.761905	21571.428571	4.142857	6.285714
std	7.783988	17562.948012	2.242448	1.707128
min	21.000000	6000.000000	1.000000	4.000000
25%	22.000000	10000.000000	2.000000	5.000000
50%	31.000000	18000.000000	3.000000	6.000000
75%	32.000000	22000.000000	6.000000	8.000000
max	45.000000	70000.000000	8.000000	9.000000

```
df.groupby('Depart')['Salary'].mean()
```

```
Depart
admin      16000.000000
finance     32000.000000
hr          17500.000000
it          40000.000000
marketing   19666.666667
sales       14000.000000
Name: Salary, dtype: float64
```

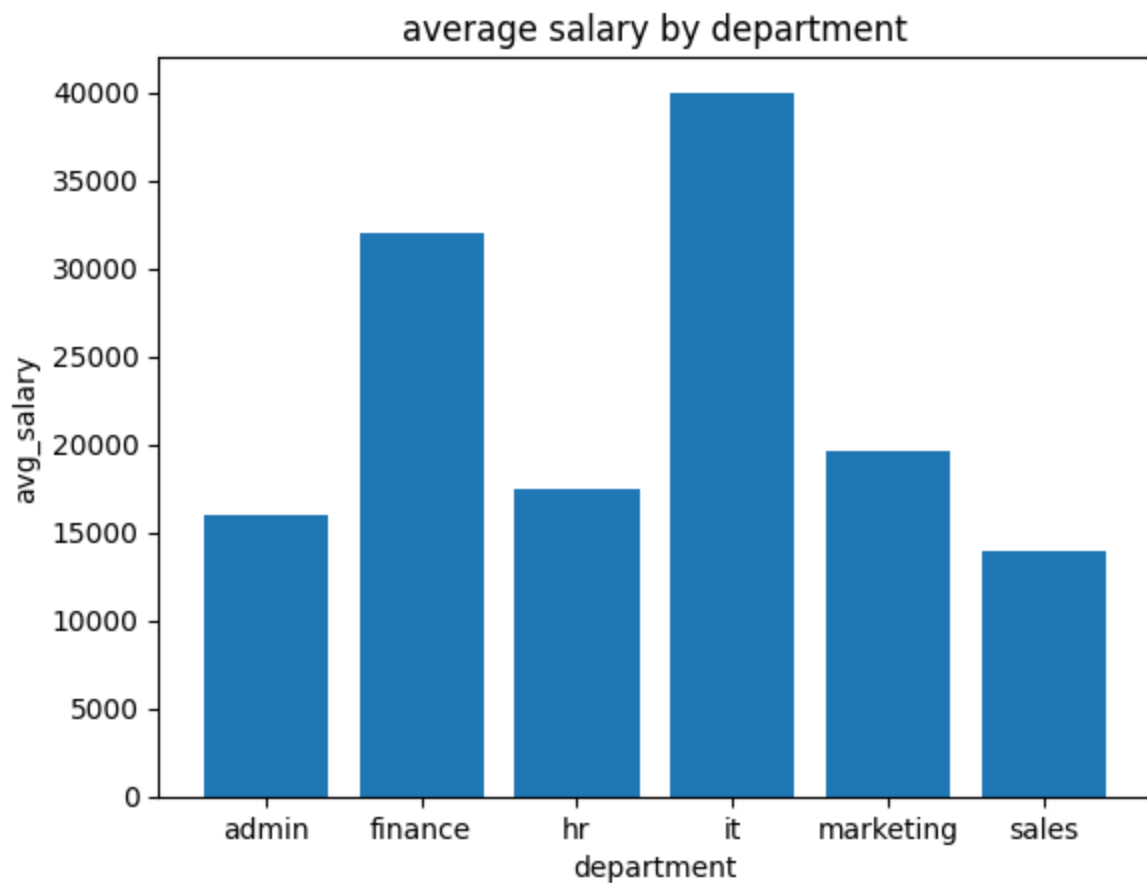
```
df.loc[df['Performance scr'].idxmax(),["name","Performance scr"]]
```

```
name      lekshmi
Performance scr      9
Name: 3, dtype: object
```

```
print(df.groupby("Depart")["Salary"].mean())
```

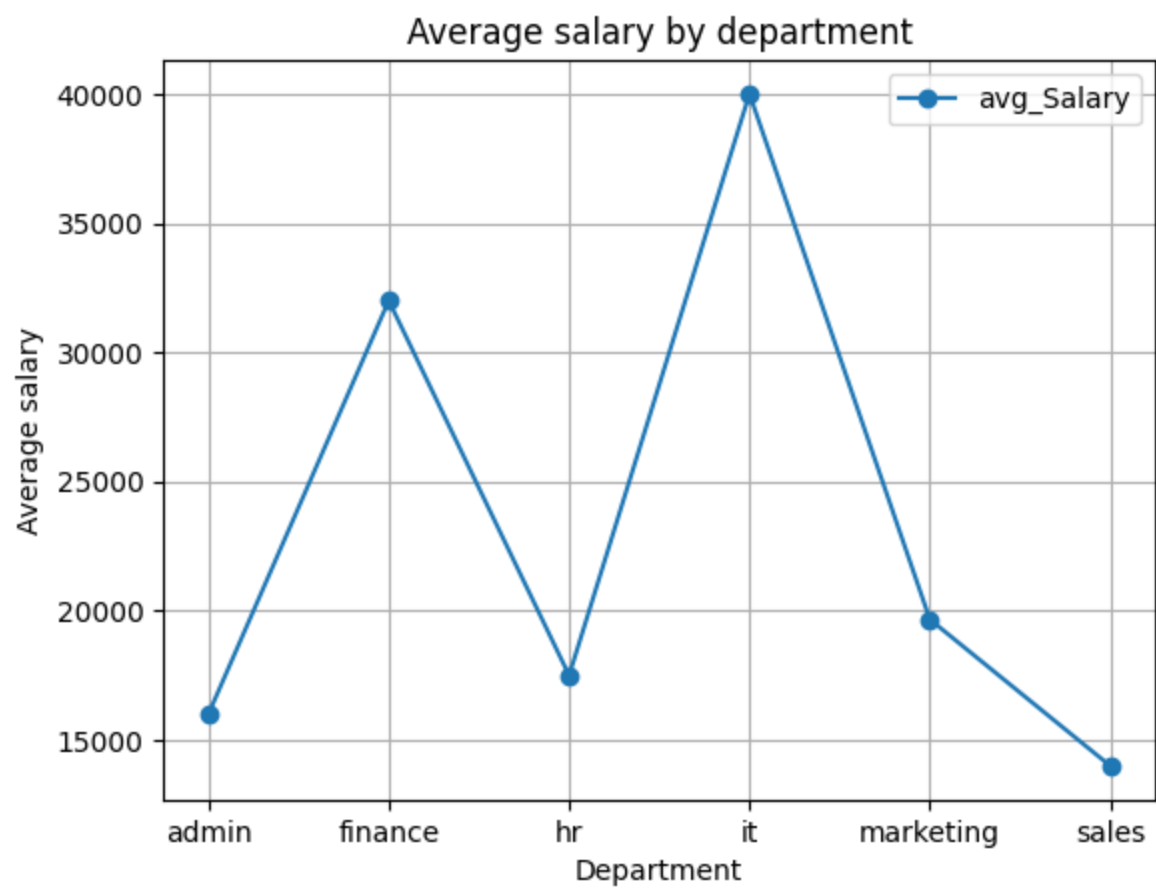
```
Depart
admin      16000.000000
finance     32000.000000
hr          17500.000000
it          40000.000000
marketing   19666.666667
sales       14000.000000
Name: Salary, dtype: float64
```

```
plt.bar(avg_Salary.index,avg_Salary)
plt.title("average salary by department")
plt.xlabel("department")
plt.ylabel("avg_salary")
plt.show()
```



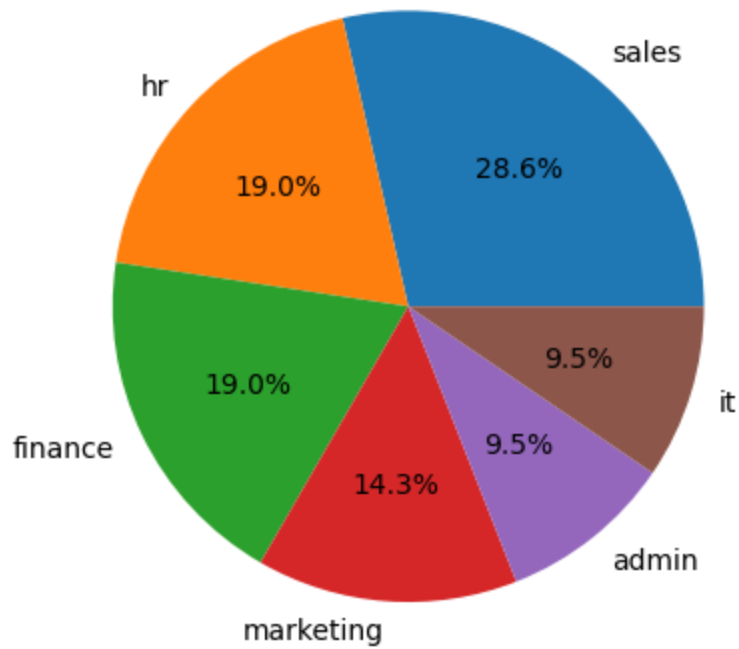
```
avg_Salary=df.groupby("Depart")["Salary"].mean()
plt.plot(avg_Salary.index,avg_Salary.values,marker="o",label="avg_Salary")
plt.title("Average salary by department")
plt.xlabel("Department")
plt.ylabel("Average salary")
plt.grid(True)
plt.legend()
```

```
plt.show()
```



```
dept_counts=df["Depart"].value_counts()
plt.pie(dept_counts.values,labels=dept_counts.index,autopct="%1.1f%%")
plt.title("Department distribution")
plt.show()
```

Department distribution



```
plt.hist(df["Salary"],bins=5,edgecolor="brown",label="Salary")
plt.title("Salary distribution")
plt.xlabel("salary")
plt.ylabel("frequency")
plt.legend()
plt.show()
```

